

Formal Modeling and Analysis of Metric-Timed Systems and Multi-Agent Normative Specifications

PhD Defense – Doctor of Natural Sciences (Dr. rer. nat)

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Outline

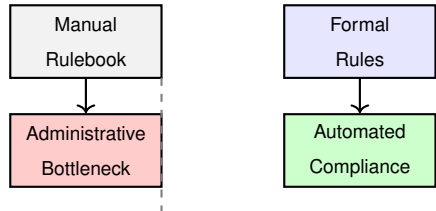
The Cost of Normative Compliance

Motivation:

- ▶ Normative systems govern behavior via **ought-to-do rules**.
- ▶ Manual verification creates a heavy economic burden.
- ▶ Real-world rules involve strict **time constraints** and **distributed agents**.

The Problem:

- ▶ Fragments of laws often **conflict** (ambiguity).
- ▶ Accountability fails when



Research Objectives

RQ1 / Chapter 3: Static Conflict Analysis

How can we formally detect, measure, and explain conflicts between time-constrained obligations and prohibitions?

RQ2 / Chapter 3: Conflict Resolution

Can we automatically repair a partially conflicting normative specification without altering its original intended behavior?

RQ3 / Chapter 4: Dynamic Blame Attribution

How do we formalize and quantify responsibility when collaborative execution of norms fails in a multi-agent setting?